Transformer-Based Sentiment Classification for Customer Service Conversations

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***Abstract*—This paper presents the result and experiments of fine-tuning transformer-based model for sentiment classification of customer service conversations. The model combines DistilBERT text encoder with categorical features (issue area, issue category, product sub-category) to predict customer sentiment. Despite severe class imbalance in training data, the model achieves 70% accuracy on a test set, with strong performance on negative and neutral classes but limited positive class prediction capability.**

I. Introduction

This work addresses a three-class sentiment classification task (negative, neutral, positive) using conversation transcripts and categorical metadata. The main challenges include: (1) class imbalance in training data, (2) long conversations requiring truncation, and (3) integrating categorical features. I propose a hybrid architecture combining fine-tuned DistilBERT with categorical features to address these challenges.

II. Dataset and Exploratory Analysis

A. Dataset Overview

The dataset contains 970 training and 30 test samples. Each sample includes conversation text and 11 categorical features. The test set is balanced, while training data exhibits extreme imbalance, with positive samples significantly underrepresented (Fig. 1). Additionally, 38% of conversations exceed the 512-token limit (mean length: 495 tokens).

Fig. 1: Sentiment distribution comparison .

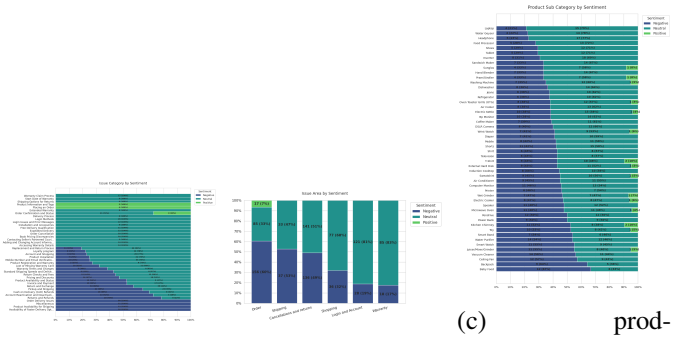
B. Data Characteristics

Feature–target analysis usung *Weighted Absolute Deviation* as metric (Table I) shows that *issue_category* and *issue_area* demonstrate the strongest relationships, while *product_sub_category* provides moderate contribution. All remaining features show small predictive value and are excluded.

TABLE I: Categorical feature relevance measured by relation score.

Feature	Relation Score	Relation Label
issue_category	0.2262	strong
issue_area	0.1693	strong
product_sub_category	0.0858	moderate
agent_experience_level	0.0313	weak
issue_sub_category	0.0302	weak
product_category	0.0168	none
issue_complexity	0.0120	none

The plots of the most informative features (Fig. 2) shows that easily resolvable issues are associated with more positive sentiment, whereas complex or unresolved issues are dominated by negative feedback.



(a) issue_category (b) issue_area (c) product_sub_category

Fig. 2: Sentiment distribution across the most informative categorical features (Table I).

III. Methodology

All experiments were conducted in a Kaggle .ipynb environment with experiment tracking and hyperparameter management performed using Weights & Biases (wandb). The configuration was divided into parts: (1) preprocessing (text processing, dialog handling, truncation strategy, etc.), and (2) model/training configuration (architecture, optimization, regularization).

A. Preprocessing

Text preprocessing included lowercase conversion and whitespace normalization. Dialogues were truncated to 512 tokens. Train/validation split: 90%/10% with stratification.

Initially assumed that sentiment is primarily located at the end of conversations, so left-side truncation was evaluated. However, experiments showed that right-side truncation

significantly improves performance (Fig. 3a), indicating that sentiment is more often lay at the beginning of dialogues.

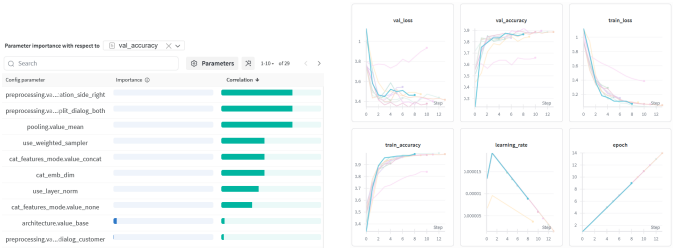
TABLE II: Mutual Information Scores for Encoding Methods

Feature	LabelEnc	OneHot	TargetEnc	Ordinal
issue_area	0.042	0.015	0.038	0.042
issue_category	0.089	0.012	0.085	0.089
product_sub_cat	0.031	0.008	0.025	0.031

Mutual Information analysis compared four encoding methods (Table II). LabelEncoder and Ordinal encoding preserved the highest mutual information with the target, while One-hot encoding was the worst and Target encoding was moderate. LabelEncoder is the safer choice because Ordinal encoding risks data leakage by using the target to define category order.

B. Model Architecture and Hyperparameter Search

The final architecture and hyperparameters were selected based on systematic experimentation on wandb (Fig. 3).



(a) Hyperparameter impact analysis. (b) Training and validation curves of the selected model.

Fig. 3: Results of wandb experiments

Text Encoder: DistilBERT with **mean pooling** over token embeddings (768-dim output), consistently outperforming CLS and max pooling.

Feature Integration: Categorical features (*issue_area*, *issue_category*, *product_sub_category*) concatenated with text embeddings (**concat mode**, 771-dim input), outperforming embedding-based and text-only baselines.

Classification Head: LayerNorm → Dropout(0.3) → Linear(771→256) → GELU → Dropout(0.3) → Linear(256→3).

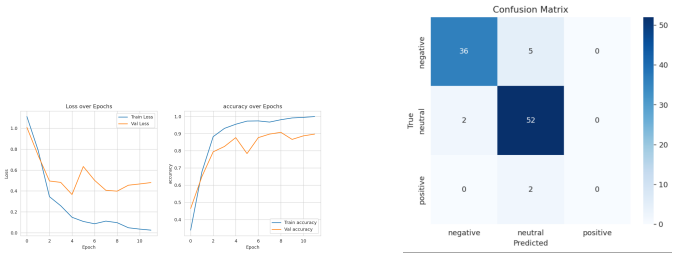
Hyperparameters highlighted as best: Pooling: **mean** (CLS, max tested); Regularization: dropout 0.3, weight decay 0.01, gradient clipping 1.0; Learning Rate: **2e-5** (1e-5 tested); Class Imbalance Handling: **WeightedRandomSampler** (weighted loss and standard training tested); Architecture Variants: additional layer and LayerNorm tested.

Final hyperparameters were chosen based on validation accuracy, complemented by F1-score and confusion matrix analysis to ensure prediction of all classes. Stable validation loss dynamics (Fig. 3b) were also considered.

IV. Results

A. Training Progress and Validation of Best Model

Training dynamics are summarized in Fig. 4a. Rapid improvement is observed in the first 5 epochs, with validation accuracy peaking at 0.9072 and early stopping triggered at epoch 12. Validation confusion matrix (Fig. 4b) shows the model predicts negative and neutral classes reliably but fails to predict the positive class due to extreme imbalance.

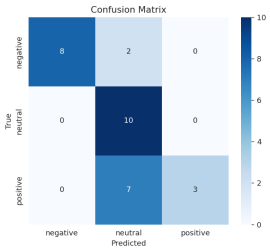


(a) Loss and accuracy over epochs. (b) Validation confusion matrix.

Fig. 4: Training progress and validation of best model.

B. Test Performance

Evaluation on the balanced test set shows overall accuracy of 0.70 and highlights differences in class-wise performance: negative and neutral are predicted reliably, while positive predictions are conservative (high precision, low recall).



(a) Confusion matrix.

Class	Precision	Recall	F1
Negative	1.00	0.80	0.89
Neutral	0.53	1.00	0.69
Positive	1.00	0.30	0.46
Macro Avg	0.84	0.70	0.68
Accuracy		0.70 (21/30)	

(b) Class-wise evaluation metrics - precision, recall, and F1-score for each class.

Fig. 5: Balanced test set evaluation.

V. Discussion

A. Key Findings

Experiments highlighted that mean pooling, direct concatenation of categorical features, LayerNorm, dropout, and WeightedRandomSampler provided the most stable and effective performance. Class imbalance still limits learning for the positive class.

B. Limitations

Small test set and extreme class imbalance reduce reliability at positive class prediction. Text truncation removes context in some conversations. Advanced techniques like data augmentation were not explored.

VI. Conclusion

A DistilBERT-based classifier combining with categorical features achieved 70% test accuracy. Negative and neutral classes are predicted well; positive class remains challenging due to imbalance. Results underline the value of careful preprocessing and systematic architecture search.